

Homework — 2.2.2 Computational Methods — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] — Two correct rows per mark:

Scenario	Method
Heat map of temperature readings	Visualisation (visualisation to solve problems)
Loyalty-card records analysed for unexpected patterns	Data mining
Rule of thumb producing a good-enough route quickly	Heuristics
Modelling how a design copes with expected peak load	Performance modelling

2. [2] — Two correct matches per mark:

Term	Match
Decomposition	C
Divide and conquer	D
Abstraction	A
Problem recognition	B

Distractors not used: E (heuristic) and F (data mining).

3.

- (a) [2] — Two correct rows per mark:

Time step	Resize	Apply filter	Add watermark
1	P1	—	—
2	P2	P1	—
3	P3	P2	P1
4	—	P3	P2
5	—	—	P3

Each photo moves one stage to the right per time step; a new photo enters Resize as soon as it is free.

- (b) [1] — From time step 3 the stages work on **different photos at the same time**, so no stage sits idle waiting for a whole photo to finish all three stages — photos are completed at a faster **rate** (one per time step once the pipeline is full).

4.

- (a) [1] — **Backtracking**.
- (b) [2] — All five stages correct = 2 marks; three or four correct = 1 mark:

Stage	Order
Reach a dead end	3
Choose a path at the first junction	1
Try a different, untried path from that junction	5
Continue extending the path while moves are available	2
Undo the most recent choice and return to the last junction with an untried option	4

5.

- (a) [1] — A **heuristic** is a rule of thumb that produces a **good-enough** answer quickly, without guaranteeing the optimal answer.
- (b) [2] — 1 mark: an exact method must consider an enormous number of possible route orderings — the number of combinations grows so quickly with the number of stops that the problem becomes **intractable** (cannot be solved exactly in a reasonable time). 1 mark: a heuristic finds a **good-enough** route **quickly**, and a near-optimal route is acceptable in practice.

6. [1] — Any one: it can be **decomposed** into smaller sub-problems; it has identifiable **inputs, processes and outputs**; it can be solved by a **finite sequence of steps** (an algorithm); it is rule-based/repeatable; the data involved can be **modelled/structured**; useful **abstractions** can be made; it is tractable — solvable in a reasonable time.

Part B (6 marks)

7. [1] — A **function returns a value** (so a call can be used in an expression); a procedure does not return a value.

8. [2] — 1 mark: a **stack overflow**. 1 mark: with no reachable base case the subroutine keeps calling itself; each call pushes a new frame (return address/parameters/locals) onto the **call stack** until the stack's memory is exhausted.

9. [3] — 1 mark: **thinking ahead** is identifying in advance what a solution will need — its inputs and outputs, preconditions, and reusable components — before/while designing it. 1 mark per distinct example, max 2: deciding what data a form must capture before building it; **caching** data/resources likely to be needed again; specifying **preconditions** for a subroutine; planning reusable components/subroutines.

Total: 14 + 6 = 20 marks